

Composite Learning Variable Impedance Robot Control With Stability and Passivity Guarantees

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Abstract—Variable impedance control (VIC) is paramount for robots to improve safety and effectiveness in physical human-robot interaction. However, achieving variable target impedance with guaranteed stability is not trivial, particularly under parametric uncertainty in the robot dynamics. This letter proposes a composite learning-based VIC (CL-VIC) strategy with three control modes, including robot-dominant, human-dominant, and collaboration modes, to achieve target impedance under parametric uncertainty. A multi-mode adaptive control scheme is defined by well-designed weighting functions to ensure smooth mode transitions. A composite learning mechanism is employed for exact parameter estimation such that target impedance models can be achieved under interval excitation that is much weaker than persistent excitation, where the latter is a sufficient condition for exact parameter estimation in classical system identification. The proposed method can achieve variable active stiffness and damping with guaranteed exponential stability and passivity of the closed-loop system, which ensures safe interaction. Experiments on a collaborative robot with seven degrees of freedom named Franka Emika Panda have validated the effectiveness of the proposed CL-VIC method.

Index Terms—Adaptive control, compliance and impedance control, parameter convergence, robot safety, physical human-robot interaction.

I. INTRODUCTION

IMPEDANCE control is an indirect force control approach that shapes the mechanical impedance of robotic systems, and it is desirable for robots to ensure safe interaction while in contact with different environments [1]. There are two methods of implementing impedance control [2]: Force-controlled impedance control and position-controlled impedance control (also called admittance control). In many scenarios, impedance

control with fixed impedance parameters does not meet task requirements [3]. Variable impedance control (VIC) not only allows controlling the dynamic relationship between external forces and robot motions but also gives the flexibility to change these dynamics continuously during task execution [4].

Impedance variation methods include impedance learning and impedance adaptation [3]. Impedance learning varies impedance parameters using optimal control, imitation learning, iterative learning, or reinforcement learning [5]. Stability is a critical problem for impedance learning, and some efforts have been made to address this problem (e.g., see [6], [7], [8]). However, impedance learning methods usually require a sizeable amount of iterative updates to obtain optimal impedance parameters. Impedance adaptation utilizes adaptive control laws rather than complex optimization algorithms to adjust impedance parameters. A common solution to achieve impedance adaptation is to build a prescribed reference human-robot interaction (HRI) model with uncertain parameters and approximate this model by tuning these impedance parameters [9], [10], [11]. Impedance adaptation usually has rigorous stability analysis but depends on the modeling of external environments (including humans).

An intuitive way for VIC is to map target impedance parameters as a specially designed function of measurable signals from robots and/or external environments such that iterative updates in impedance adaptation and learning become unnecessary. Such an attractive feature has made VIC with direct adjustment mechanisms popular in recent developments, where one of its crucial aspects is a stability guarantee. An intuitive approach to guarantee stability is restricting variable impedance parameters to their stable regions determined by experiments rather than rigorous theoretical analysis such that environment modeling can be avoided [12], [13], [14]. Yet, obtaining effective stable regions requires numerous trials, and such regions only work in similar experimental environments. Frequency response analysis in classical control theory has been applied to evaluate stable regions of impedance parameters [15], [16], [17]. Critical impedance parameters were derived by virtue of Lyapunov theory in [18]. In the above-mentioned VIC methods, determining the stable regions or critical values of impedance parameters requires prior knowledge of environmental models.

Passivity can guarantee safety when robots are in contact with passive environments without prior knowledge of environmental models [19]. One way to ensure passivity in VIC with direct adjustment is to monitor and control the power/energy injected into the system. In [19], [20], [21], [22], a virtual energy tank

Manuscript received 17 June 2023; accepted 8 October 2023. Date of publication 25 October 2023; date of current version 21 November 2023. This letter was recommended for publication by Associate Editor F. Pierri and Editor Clement G. upon evaluation of the reviewers' comments. This work was supported in part by the Guangdong Provincial Pearl River Talents Program, China, under Grant 2019QN01X154, and in part by the Fundamental Research Funds for the Central Universities, Sun Yat-sen University, China, under Grant 23lgzy004. (Corresponding author: Yongping Pan.)

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This letter has supplementary downloadable material available at <https://doi.org/10.1109/LRA.2023.3327675>, provided by the authors.

Digital Object Identifier 10.1109/LRA.2023.3327675

acting as an energy reservoir was designed to track the energy dissipated by the system so that impedance parameters can vary under preserved passivity. In [23], a valve-based virtual energy tank was applied to multi-manual object manipulation tasks, where the power flows between the tank and system were also controlled in addition to tank energy, such that this tank can handle several potentially non-passive ports based on task-execution hierarchy. In [24], potential energy storage was applied to vary stiffness, providing a clearer physical interpretation than the virtual energy tank. A major drawback of energy tank-based VIC approaches is that the lower and upper bounds of the energy stored in the tank should be determined previously for a specific task, making them impractical for some complex situations. A power envelope to limit the power injected into the human-robot coupled system was developed in [25], where the bounds of the tank energy are not required in advance for VIC.

Another way to ensure passivity in VIC is to apply passivity conditions directly, which does not require tracking dissipative power or energy online [26], [27], [28], [29], [30], [31], [32]. A passivity filter was designed in [26] to modify the non-passive impedance profile in a way that guarantees passivity, where it is allowed to be used either offline or online. In [27], a passivity preservation method was applied to VIC of flexible-joint robots (FJR) for safe impedance variation in regulation tasks. This method was extended to tracking tasks of FJR in [28]. In [29], a passive VIC method was developed for humanoid robots with series elastic actuators (SEAs). In some scenarios, the robot needs to change working modes to adapt its task stages. A passive VIC method with variable stiffness was proposed in [30] for SEA-driven robots to guarantee stable and smooth HRI under robot-in-charge and human-in-charge modes. This work was extended to design a passive VIC method with variable damping for HRI with augmentation and autonomous modes in [31], and also extended to a human intention-based VIC method with assistance and passive modes for SEA-driven robots in [32].

To achieve target impedance models, the above-mentioned VIC methods depend on the assumption of perfect inner-loop position control or on exact robot modeling with only a few exceptions resorting to adaptive control as in [30], [31]. However, a stringent condition termed persistent excitation (PE) must be fulfilled to realize target impedance in [30], [31]. On the other hand, in contrast to position-controlled impedance control (e.g., [6], [31]), force-controlled impedance control can work at low impedance parameters [33], such that it is stable even for very stiff environments [2]. But in this case, it suffers from poor tracking accuracy in free motion [33].

Motivated by the above fact, this letter proposes a composite learning-based VIC (CL-VIC) strategy that belongs to force-controlled impedance control. A composite learning mechanism is developed for exact robot modeling online, such that the target impedance can be easier to achieve under interval excitation (IE) that only requires rich data within a certain period, as compared to PE, which requires persistent and rich data at all times. A multi-mode adaptive control scheme with three control modes, including robot-dominant, human-dominant, and collaboration modes, is defined based on weighting functions to guarantee smooth mode transitions. The proposed method can achieve: 1)

variable target stiffness and damping to adapt to different contact environments while ensuring exponential stability and passivity of the closed-loop system, and 2) high-accuracy tracking under low impedance parameters, attributed to the exact estimation of the unknown robot dynamics. Experiments are conducted on a collaborative robot with 7 degrees of freedom (DoF) to validate the effectiveness of the proposed CL-VIC method.

In this letter, $\mathbb{R}$, $\mathbb{R}^+$, $\mathbb{R}^n$ and $\mathbb{R}^{m \times n}$ represent the spaces of real numbers, positive real numbers, real n -vectors and real $m \times n$ -matrices, respectively, $\mathbb{N}$ denotes a set of natural numbers, $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ are the minimum and maximum eigenvalues of $A \in \mathbb{R}^{n \times n}$, respectively, $\min\{\cdot\}$ and $\max\{\cdot\}$ are the minimum and maximum operators, respectively, L_∞ is the space of bounded signals, $\|\mathbf{x}\|$ is the Euclidean norm of $\mathbf{x} \in \mathbb{R}^n$, and $\text{diag}(x_1, x_2, \dots, x_n)$ is a diagonal matrix with diagonal elements $x_i \in \mathbb{R}$ with $i = 1$ to n , where $n, m \in \mathbb{N}$.

II. PROBLEM FORMULATION

Consider a class of n -DoF robots described by the following Euler-Lagrange formulation [34]:

$$M(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + F(\dot{\mathbf{q}}) + G(\mathbf{q}) = \boldsymbol{\tau} + \boldsymbol{\tau}_e \quad (1)$$

in which $\mathbf{q}(t) = [q_1(t), q_2(t), \dots, q_n(t)]^T \in \mathbb{R}^n$ denotes a joint angular position vector, $M(\mathbf{q}) \in \mathbb{R}^{n \times n}$ is an inertia matrix, $C(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{n \times n}$ is a matrix of centripetal-Coriolis terms, and $F(\dot{\mathbf{q}}) \in \mathbb{R}^n$, $G(\mathbf{q}) \in \mathbb{R}^n$, $\boldsymbol{\tau} \in \mathbb{R}^n$, and $\boldsymbol{\tau}_e \in \mathbb{R}^n$ are friction, gravity, control, and contact torque vectors, respectively. It is assumed that $\mathbf{q}$ and $\dot{\mathbf{q}}$ are measurable, and $\ddot{\mathbf{q}}$ is not accessible for measurement. The following properties and definitions are presented to facilitate control design [35].

Property 1: $M(\mathbf{q})$ is symmetric and positive-definite, and satisfies $m_0\|\mathbf{x}\|^2 \leq \mathbf{x}^T M(\mathbf{q})\mathbf{x} \leq \bar{m}(\mathbf{q})\|\mathbf{x}\|^2$, $\forall \mathbf{x} \in \mathbb{R}^n$, with $\bar{m} : \mathbb{R}^n \mapsto \mathbb{R}^+$ and $m_0 \in \mathbb{R}^+$ being a constant.

Property 2: $\dot{M}(\mathbf{q}) - 2C(\mathbf{q}, \dot{\mathbf{q}})$ is skew-symmetric such that $\mathbf{x}^T (\dot{M}(\mathbf{q}) - 2C(\mathbf{q}, \dot{\mathbf{q}}))\mathbf{x} = 0$, $\forall \mathbf{x} \in \mathbb{R}^n$, which implies that the internal forces do not work.

Property 3: The left-hand side of the robotic system (1) can be linearly parameterized as follows:

$$G(\mathbf{q}) + F(\dot{\mathbf{q}}) + C(\mathbf{q}, \dot{\mathbf{q}})\mathbf{v} + M(\mathbf{q})\dot{\mathbf{v}} = \Phi^T(\mathbf{q}, \dot{\mathbf{q}}, \mathbf{v}, \dot{\mathbf{v}}) \mathbf{W} \quad (2)$$

in which $\mathbf{v} \in \mathbb{R}^n$ is an auxiliary variable, $\Phi : \mathbb{R}^{4n} \mapsto \mathbb{R}^{N \times n}$ is a known smooth regressor, $\mathbf{W} \in \mathbb{R}^N$ is an unknown parameter vector, and N is the number of parameters.

Definition 1: A bounded signal $\Phi(t) \in \mathbb{R}^{N \times n}$ is of PE, if $\exists \sigma, \varsigma_d \in \mathbb{R}^+$ such that $\int_{t-\varsigma_d}^t \Phi(\varsigma)\Phi^T(\varsigma)d\varsigma \geq \sigma I$, $\forall t \geq 0$.

Definition 2: A bounded signal $\Phi(t) \in \mathbb{R}^{N \times n}$ is of IE at $t = t_e$, if $\exists t_e, \varsigma_d, \sigma \in \mathbb{R}^+$ so that $\int_{t_e-\varsigma_d}^{t_e} \Phi(\varsigma)\Phi^T(\varsigma)d\varsigma \geq \sigma I$.

It is clear from Definitions 1 and 2 that the PE condition is stronger as it requires that excitation exists for all time $t \geq 0$, and the IE condition is strictly weaker than the PE condition as it only requires excitation to occur at a particular time $t = t_e$ which may be fulfilled during transient processes. Let $\mathbf{q}_d(t) = [q_{d1}(t), q_{d2}(t), \dots, q_{dn}(t)]^T \in \mathbb{R}^n$ be a desired output that satisfies $\mathbf{q}_d, \dot{\mathbf{q}}_d, \ddot{\mathbf{q}}_d \in L_\infty$, and $\hat{\mathbf{W}}(t) \in \mathbb{R}^N$ be an estimate of $\mathbf{W}$. Define a tracking error $\mathbf{e}(t) := \mathbf{q}_d(t) - \mathbf{q}(t)$, an estimation error $\tilde{\mathbf{W}}(t) := \mathbf{W} - \hat{\mathbf{W}}(t)$, and a smoothed contact torque $\boldsymbol{\tau}_{es}(t) :=$

$L_1(s)\{\tau_e(t)\}$ with $L_1(s) := \frac{\lambda_1}{s+\lambda_1}$, where $\lambda_1 \in \mathbb{R}^+$ is a filtering constant, and s is the complex Laplace operator. Note that τ_{es} is applied to exclude high-frequency noise and to avoid abrupt changes. We aim to develop a suitable control strategy for the robotic system (1) with Properties 1–3 to achieve exact tracking and safe interaction with guaranteed dynamic stability.

III. HUMAN-ROBOT INTERACTION CONTROL

This section introduces a multi-mode adaptive robot control scheme. Firstly, a multi-mode VIC controller with an estimated value $\hat{W}$ is designed in Section III-A. After that, a state-of-the-art composite learning technique is utilized to realize efficient and precise online estimation of the unknown parameter W by $\hat{W}$ in Section III-B, where the closed-loop system is proven stable and passive rigorously. Finally, detailed explanations about different working modes are given in Section III-C.

A. Multi-Mode Control Scheme

To design a multi-mode controller, define

$$\omega(x, a, b) := \begin{cases} 1, & x \in [b, \infty) \\ \frac{1}{2} + \frac{1}{2} \sin(\frac{\pi}{\sigma}(x - b) + \frac{\pi}{2}), & x \in (a, b) \\ 0, & x \in [0, a] \end{cases} \quad (3)$$

as a derivable weighting function, where $x(t) \in \mathbb{R}^+$ is a time-dependent variable, $a, b \in \mathbb{R}^+$ ($a \geq b$) denote two prescribed boundary constants, and $\sigma := b - a \in \mathbb{R}^+$ denotes the width of $\omega \in (0, 1)$. The time derivative of ω is

$$\dot{\omega}(x, a, b) = \begin{cases} \frac{\pi \dot{x}}{2\sigma} \cos(\frac{\pi}{\sigma}(x - b) + \frac{\pi}{2}), & x \in (a, b) \\ 0, & \text{others} \end{cases} \quad (4)$$

Define a weighted error $z = Q[\tau_{es}^T, e^T]^T \in \mathbb{R}^{2n}$, where $Q \in \mathbb{R}^{2n \times 2n}$ ($Q > 0$) is a diagonal weighting matrix of the smoothed contact torque τ_{es} and the tracking error e . In classical robot motion control, a filtered tracking error $e_s(t) := \dot{e}(t) + \Lambda e(t) \in \mathbb{R}^n$ is defined to achieve trajectory tracking, where $\Lambda \in \mathbb{R}^{n \times n}$ is a positive-definite diagonal matrix. Then, one has $e_s(t) = \dot{q}_r(t) - \dot{q}(t)$ with $\dot{q}_r(t) := \dot{q}_d(t) + \Lambda e(t)$ being a “reference velocity”. To achieve the above three modes, redefine

$$\dot{q}_r := (1 - \omega_s(t))(\dot{q}_d + \Lambda e) \quad (5)$$

in which $\omega_s(t) := \omega(\|z\|, \delta_1, \delta_2)$ is a weighting function for stiffness variation with $\delta_2, \delta_1 \in \mathbb{R}^+$. It follows from (5) that

$$\ddot{q}_r = (1 - \omega_s(t))(\ddot{q}_d + \Lambda \dot{e}) - \dot{\omega}_s(t)(\dot{q}_d + \Lambda e), \quad (6)$$

where $\dot{\omega}_s(t)$ can be calculated by substituting $a = \delta_1$, $b = \delta_2$, and $\dot{x} = \frac{d\|z\|}{dt} = \frac{z^T \dot{z}}{\|z\|}$ into (4).

Differentiating e_s with respect to t and using (1) yields

$$M(q)\dot{e}_s = M(q)\ddot{q}_r + C(q, \dot{q})\dot{q} + F(\dot{q}) + G(q) - \tau_e - \tau.$$

Applying $\dot{q} = \dot{q}_r - e_s$ to the above equality leads to

$$M(q)\dot{e}_s = \Phi^T(q, \dot{q}, \ddot{q}_r, \ddot{q}_r)W - C(q, \dot{q})e_s - \tau_e - \tau. \quad (7)$$

Design a multi-mode control law as follows:

$$\tau = \omega_d(\dot{q}, \tau_{es})K_c e_s + \Phi^T(q, \dot{q}, \ddot{q}_r, \ddot{q}_r)\hat{W} \quad (8)$$

where $K_c \in \mathbb{R}^{n \times n}$ ($K_c > 0$) is a diagonal gain matrix, and

$$\omega_d(\dot{q}, \tau_f) = \max\{\alpha_2 e^{-\beta\omega(\|\tau_{es}\|, \delta_3, \delta_4)\|\dot{q}\|}, \alpha_1\} \quad (9)$$

with $\alpha_1, \alpha_2, \beta, \delta_4, \delta_3 \in \mathbb{R}^+$ is a weighting function for damping variation. Note that $\ddot{q}_r$ given by (6) in (8) is useful to maintain stability and passivity of the closed-loop system under variable impedance. Applying (8) to (7), one obtains the closed-loop tracking error dynamics as follows:

$$M(q)\dot{e}_s = -\omega_d K_c e_s - C(q, \dot{q})e_s + \Phi_r^T \tilde{W} - \tau_e \quad (10)$$

with $\Phi_r := \Phi(q, \dot{q}, \ddot{q}_r, \ddot{q}_r)$ for short.

B. Composite Learning Interaction Control

Applying (2) with $v = \dot{q}$ to (1), one obtains

$$\tau + \tau_e = \Phi^T(q, \dot{q}, \ddot{q})W. \quad (11)$$

To avoid the appearance of the unavailable $\ddot{q}$, filtering each side of (11) by $L_2(s) := \frac{\lambda_2}{s+\lambda_2}$ results in

$$\tau_f(t) + \tau_{ef}(t) = \Phi_f^T(q, \dot{q})W \quad (12)$$

with $\lambda_2 \in \mathbb{R}^+$ being a filtering constant, $\tau_f := L_2(s)\{\tau\}$, $\tau_{ef} := L_2(s)\{\tau_e\}$, and $\Phi_f(q, \dot{q}) := L_2(s)\{\Phi(q, \dot{q}, \ddot{q})\}$. Define

$$\Theta(t) := \int_{t-\varsigma_d}^t \Phi_f(\varsigma) \Phi_f^T(\varsigma) d\varsigma \quad (13)$$

with $\Phi_f(t) := \Phi_f(q, \dot{q})$ for short. Then, multiplying (12) by Φ_f , integrating the result at $[t - \varsigma_d, t]$ and using (13), one gets

$$\Theta(t)W = \int_{t-\varsigma_d}^t \Phi_f(\varsigma)(\tau_f(\varsigma) + \tau_{ef}(\varsigma))d\varsigma := \varphi(t). \quad (14)$$

To take full advantage of excitation information for parameter estimation, define a torque prediction error

$$\epsilon(t) := \tau_f(t) + \tau_{ef}(t) - \Phi_f^T(q, \dot{q})\hat{W}(t) \quad (15)$$

and a generalized torque prediction error

$$\xi(t) := \begin{cases} \varphi(t) - \Theta(t)\hat{W}(t), & t < t_e \\ \varphi(t_e) - \Theta(t_e)\hat{W}(t), & t \geq t_e \end{cases} \quad (16)$$

where $(\tau_f + \tau_{ef})$ and φ satisfy (12) and (14), respectively, and φ is calculable by the integral part of (14). Design a composite learning update law for $\hat{W}$ as follows:

$$\dot{\hat{W}} = \Gamma(k_a \Phi_f(q, \dot{q})\epsilon + k_b \xi) \quad (17)$$

in which $k_a, k_b \in \mathbb{R}^+$ are weighting factors, and $\Gamma \in \mathbb{R}^{N \times N}$ is a positive-definite diagonal matrix of learning rates. Fig. 1 shows the proposed control strategy for human-robot-environment interaction, where (17) is a self-tuning estimator to update $\hat{W}$ without using the filtered tracking error e_s as it may destroy parameter convergence during interaction. The theoretical results of this study are presented as follows.

Theorem 1: If the robotic system (1) is driven by the control law (8) with (17) under the IE condition in Definition 2, the closed-loop system achieves: 1) stability in the sense that both the estimation error $\tilde{W}$ and the tracking error e exponentially

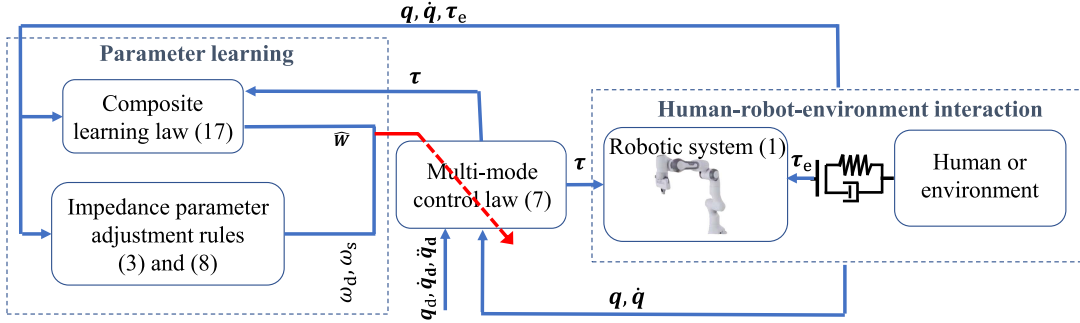


Fig. 1. Block diagram of the proposed control strategy for human-robot-environment interaction.

converge to $\mathbf{0}$ when the contact torque $\tau_e = \mathbf{0}$, and 2) passivity regarding the input τ_e and the output e_s .

Proof: Choose a Lyapunov function candidate

$$V = e_s^T M(q) e_s / 2 + \underbrace{\tilde{W}^T \Gamma^{-1} \tilde{W} / 2}_{V_w} \quad (18)$$

for the closed-loop system (10) with (17). Differentiating V_w in (18) with respect to t and applying (17) produces

$$\dot{V}_w = -\kappa_a \tilde{W}^T \Phi_f \Phi_f^T \tilde{W} - \kappa_b \tilde{W}^T \Theta(t) \tilde{W} \leq 0 \quad (19)$$

which verifies $0 \leq V_w(t) \leq V_w(0)$, $\forall t \geq 0$ implying $\tilde{W}, \dot{\tilde{W}} \in L_\infty$. If IE holds (i.e., $\Theta(t) = \Theta(t_e) \geq \sigma I$), from (19) one has

$$\dot{V}_w \leq -(2\kappa_b \sigma / \lambda_{\max}(\Gamma^{-1})) V_w$$

which verifies the exponential convergence of $\tilde{W}$ to $\mathbf{0}$.

Differentiating V in (18) with respect to t , one gets

$$\dot{V} = e_s^T M(q) \dot{e}_s + e_s^T M(q) e_s / 2 + \dot{V}_w.$$

Applying (10) and (19) to the above equality and noting the skew-symmetry in *Property 2*, one obtains

$$\begin{aligned} \dot{V} = & -\omega_d(\dot{q}) e_s^T K_c e_s - e_s^T \tau_e + e_s^T \Phi_r^T \tilde{W} \\ & - \kappa_a \tilde{W}^T \Phi_f \Phi_f^T \tilde{W} - \kappa_b \tilde{W}^T \Theta(t) \tilde{W}. \end{aligned} \quad (20)$$

According to Young's inequality, one has

$$a^T b \leq \frac{1}{2\mu} a^T a + \frac{\mu}{2} b^T b, \forall a, b \in \mathbb{R}^n, \mu \in \mathbb{R}^+. \quad (21)$$

Substituting $a = -e_s$ and $b = \tau_e$ into (21) yields

$$-e_s^T \tau_e \leq \mu e_s^T e_s / 2 + \|\tau_e\|^2 / (2\mu).$$

Substituting $a = e_s$ and $b = \Phi_r^T \tilde{W}$ into (21) yields

$$\begin{aligned} e_s^T \Phi_r^T \tilde{W} & \leq \mu e_s^T e_s / 2 + (\Phi_r^T \tilde{W})^T \Phi_r^T \tilde{W} / (2\mu) \\ & = \mu e_s^T e_s / 2 + \tilde{W}^T \Phi_r \Phi_r^T \tilde{W} / (2\mu). \end{aligned}$$

Applying these inequalities to (20) results in

$$\begin{aligned} \dot{V} \leq & -\omega_d(\dot{q}) e_s^T K_c e_s - \kappa_b \tilde{W}^T \Theta(t) \tilde{W} + \mu e_s^T e_s \\ & + \underbrace{(\tilde{W}^T \Phi_r \Phi_r^T \tilde{W} / (2\mu) - \kappa_a \tilde{W}^T \Phi_f \Phi_f^T \tilde{W})}_{\delta_\phi} + \frac{\|\tau_e\|^2}{2\mu}. \end{aligned}$$

The exponential convergence of $\tilde{W}$ to $\mathbf{0}$ implies $\delta_\phi \in L_\infty$ and its exponential convergence to $\mathbf{0}$. Once IE in Definition 2 holds,

one gets $\Theta(t) \geq \sigma I$ such that $\tilde{W}^T \Theta \tilde{W} \geq \sigma \tilde{W}^T \tilde{W}$. Besides, K_c is a positive-definite diagonal matrix in (8), such that $e_s^T K_c e_s \geq \lambda_{\min}(K_c) e_s^T e_s$. Consequently, one gets

$$\begin{aligned} \dot{V} \leq & -(\omega_d(\dot{q}) \lambda_{\min}(K_c) - \mu) e_s^T e_s \\ & - \kappa_b \sigma \tilde{W}^T \tilde{W} + \|\tau_e\|^2 / (2\mu) + \delta_\phi \\ = & -\frac{(\omega_d(\dot{q}) \lambda_{\min}(K_c) - \mu)}{\lambda_{\max}(M(q))} \lambda_{\max}(M(q)) e_s^T e_s \\ & - \frac{\kappa_b \sigma}{\lambda_{\max}(\Gamma^{-1})} \lambda_{\max}(\Gamma^{-1}) \tilde{W}^T \tilde{W} + \|\tau_e\|^2 / (2\mu) + \delta_\phi. \end{aligned}$$

Properly choosing K_c and μ can ensure $\omega_d(\dot{q}) \lambda_{\min}(K_c) - \mu > 0$ with $\omega_d \geq \alpha_1 > 0$. As $e_s^T M(q) e_s \leq \lambda_{\max}(M(q)) e_s^T e_s$ and $\tilde{W}^T \Gamma^{-1} \tilde{W} \leq \lambda_{\max}(\Gamma^{-1}) \tilde{W}^T \tilde{W}$, one gets

$$\begin{aligned} \dot{V} \leq & -\frac{(\omega_d(\dot{q}) \lambda_{\min}(K_c) - \mu)}{\lambda_{\max}(M(q))} e_s^T M(q) e_s \\ & - \frac{\kappa_b \sigma}{\lambda_{\max}(\Gamma^{-1})} \tilde{W}^T \Gamma^{-1} \tilde{W} + \|\tau_e\|^2 / (2\mu) + \delta_\phi \\ \leq & -\rho V + \|\tau_e\|^2 / (2\mu) + \delta_\phi \end{aligned} \quad (22)$$

with $\rho := \min(\frac{(\omega_d(\dot{q}) \lambda_{\min}(K_c) - \mu)}{\lambda_{\max}(M(q))}, \frac{\kappa_b \sigma}{\lambda_{\max}(\Gamma^{-1})}) \in \mathbb{R}^+$, which implies that the closed-loop system is stable in the sense of $e_s, q, \dot{q} \in L_\infty$ as long as $\tau_e \in L_\infty$. When $\tau_e = \mathbf{0}$, one obtains

$$V \leq e^{-\rho t} V(0) + \int_0^t e^{-\rho(t-\varsigma)} \delta_\phi(\varsigma) d\varsigma.$$

As δ_ϕ is exponentially dissipated, the above result implies that e_s exponentially converges to $\mathbf{0}$. Since e can be viewed as the output of a first-order filter with e_s as an input, e exponentially converges to $\mathbf{0}$ as well.

Omitting the exponentially dissipated δ_ϕ and through similar operations to those in (22), (20) becomes

$$\begin{aligned} \dot{V} \leq & -(\omega_d(\dot{q}) \lambda_{\min}(K_c) - \mu/2) e_s^T e_s \\ & - \kappa_b \sigma \tilde{W}^T \tilde{W} - e_s^T \tau_e \\ \leq & -\varrho V - e_s^T \tau_e \end{aligned} \quad (23)$$

in which $\varrho := \min(\frac{(\omega_d(\dot{q}) \lambda_{\min}(K_c) - \mu/2)}{\lambda_{\max}(M(q))}, \frac{\kappa_b \sigma}{\lambda_{\max}(\Gamma^{-1})}) \in \mathbb{R}^+$ if μ is sufficiently small, which implies the closed-loop robot dynamics is passive between the input τ_e and the output e_s , such that the CL-VIC is robust against the contact torque τ_e . ■

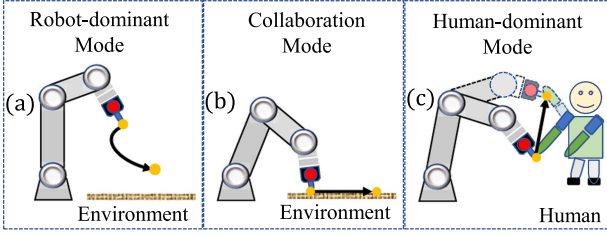


Fig. 2. Illustration of the three control modes, where the arrows denote the trajectories of the robot end-effector.

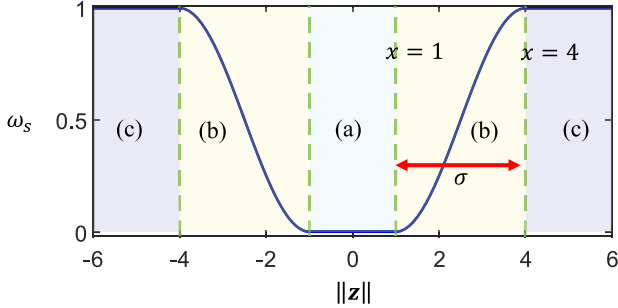


Fig. 3. Illustrating the weighting function ω_s in (5) with $\delta_1 = 1$ and $\delta_2 = 4$, where the areas (a), (b), and (c) denote the robot-dominant, collaboration, and human-dominant modes, respectively, and the width of the area (b) is $\sigma = 3$.

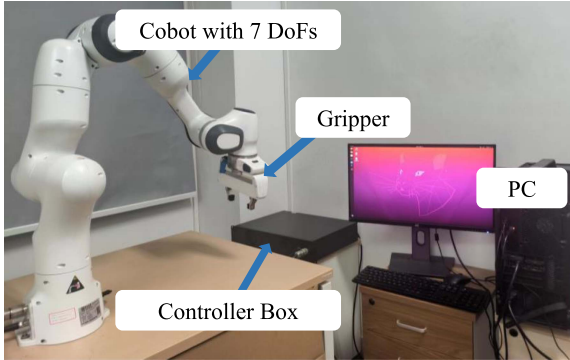


Fig. 4. An experimental setup of the collaborative robot (cobot).

C. Analysis of the Multiple Modes

When executing some complicated HRI tasks, a robot needs smooth transitions between different modes to adapt its working states, and the proposed CL-VIC exactly meets such a requirement. Besides, (22) and (23) hold regardless of the value of the weighting function ω_s , which indicates that the closed-loop system is always stable and passive, independent of the modes (i.e., ω_s). However, the robot has some special characteristics under different modes. Moreover, the IE condition would hold after a short learning phase, leading to the convergence of the term $\Phi_r \tilde{W}$ to $\mathbf{0}$ in accordance with the exponential convergence of the estimation error $\tilde{W}$. Hence, we elaborate the compliant robot behaviors in different modes under $\tilde{W} = \mathbf{0}$.

In a *robot-dominant mode*, the robot runs in a contact-free state (i.e., $\tau_e \approx 0$) [see Fig. 2(a)], where the tracking error e

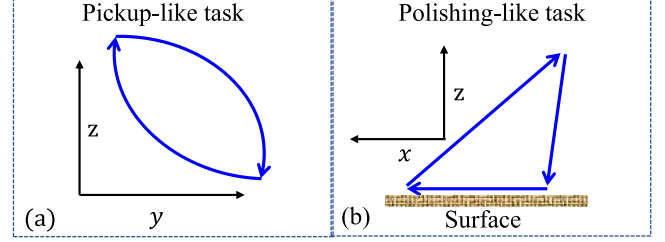


Fig. 5. Illustration of pickup and polishing-like tasks. (a) The pickup-like task (S1). (b) The polishing-like task (S3 and S4). Note that blue arrows denote the trajectories of the end-effector in each task.

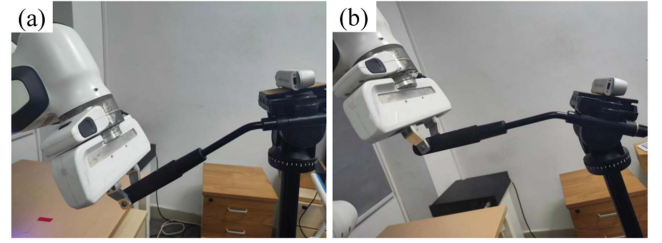


Fig. 6. Illustration of the tool operation task (S5). (a) The first camera pose. (b) The second camera pose.

exponentially converges to $\mathbf{0}$ [see Theorem 1]. Once $\|z\| \leq \delta_1$, we can get $\omega_s = 1$ and $\omega_d = \alpha_2$ in this state. Then, the closed-loop robot dynamics (10) becomes

$$M(q)\dot{e}_s + C(q, \dot{q})e_s + \alpha_2 K_c e_s = \mathbf{0}.$$

Therefore, the robot exhibits the same precise tracking performance as constant impedance control in this mode.

In a *collaboration mode*, the robot performs tasks requiring frequent contact and interaction with the environment [see Fig. 2(b)], e.g., writing, wiping, and pushing. In this mode, the closed-loop robot dynamics (10) becomes

$$M(q)\dot{e}_s + C(q, \dot{q})e_s + \omega_d(\dot{q})K_c e_s = -\tau_e$$

with $\|z\| \in (\delta_1, \delta_2)$ and $w_s \in (0, 1)$ varying with the contact torque τ_e and the tracking error e . Hence, the robot can adjust its impedance parameters to adapt to the environment with different stiffness, such that the contact torque τ_e would not be large enough to damage the contact environment. Moreover, w_s can act as a safety evaluator, as the large tracking error e and contact torque τ_e imply task failure or dangerous-impact contact.

Once the tracking error e or the contact torque τ_e increases resulting in $\|z\| \geq \delta_2$, one has $w_s = 1$, such as when the robot has a serious-impact contact or is blocked by an unmovable object, the robot enters a *human-dominant mode* such that the closed-loop robot dynamics (10) becomes

$$M(q)\ddot{q} + (C(q, \dot{q}) + \omega_d(\dot{q})K_c)\dot{q} = -\tau_e.$$

The robot has zero stiffness in this mode, which makes it completely compliant with the environment or human [see Fig. 2(c)]. Thereby, any ensuing damage can be avoided. Also, the human can actively guide the robot to enter the human-dominant mode to complete tasks like learning from demonstration.

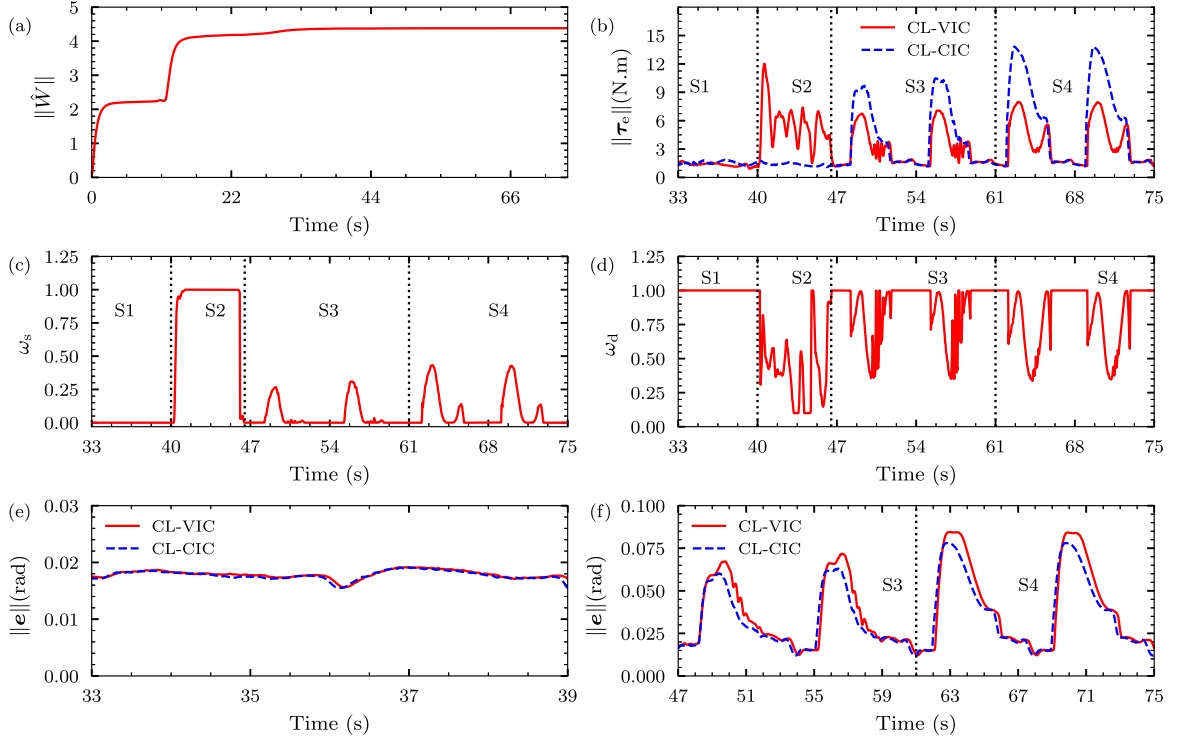


Fig. 7. Experimental results by two controllers in Tasks S1-S4. (a) The norm of the parameter estimate $\hat{W}$. (b) The norm of the contact torque τ_e . (c) The weighting function ω_s for stiffness variation. (d) The weighting function ω_d for damping variation. (e) The norms of the tracking error e in S1. (f) The norms of the tracking error e in S3-S4.

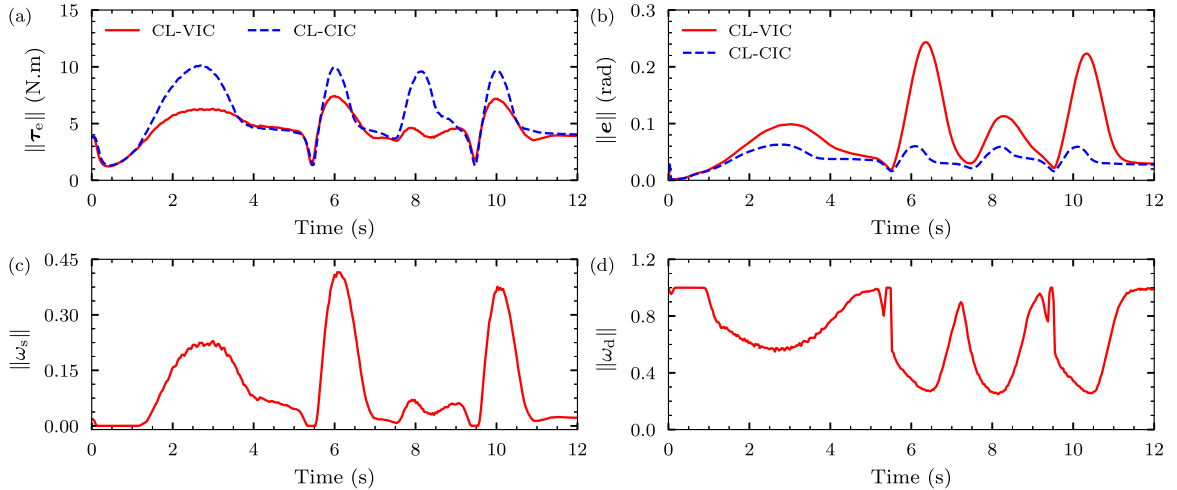


Fig. 8. Experimental results by two controllers in Task S5. (a) The norm of the contact torque τ_e . (b) The norms of the tracking error e . (c) The weighting function ω_s for stiffness variation. (d) The weighting function ω_d for damping variation.

Since the target robot impedance can vary continuously from high to low values as the weighting function $w_s \in [0, 1]$ varies [see Fig. 3], a smooth transition between the robot-dominant and human-dominant modes can be achieved to increase robot safety and efficacy. In any of the above modes, the weighting function ω_d can vary continuously within the range $[\alpha_1, \alpha_2]$ based on the robot velocity $\dot{q}$ to adjust the target damping such that robot safety and efficacy can be further increased.

IV. EXPERIMENTAL STUDIES

Experiments are performed based on the 7-DoF Panda robot to validate the proposed CL-VIC method [see Fig. 4]. Five tasks denoted by S1 to S5 are designed to exhibit the characteristics of the robot in different modes. S1 is a pickup-like task where the robot works in the robot-dominant mode [see Fig. 5(a)]. S2 is a human-robot interaction task where the robot works in

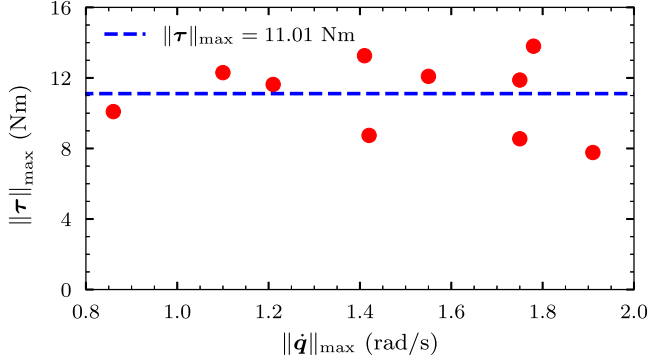


Fig. 9. Safety test results of the human-dominant mode by the proposed CL-VIC in Task S2. Note that experimental data were collected from 10 subjects.

the human-dominant mode. S3 and S4 are polishing-like tasks with soft and hard surfaces, respectively, where the robot is in collaboration mode when it comes into physical contact with the surfaces [see Fig. 5(b)]. The human intervenes and turns the robot into the human-dominant mode at the end of S1 by exerting a force to drag it away from its desired trajectory. In S2, the human guides the robot near the initial position of S3 and exits at the end of S2. Thus, the robot will autonomously turn into robot-dominant mode before touching the surfaces in S3. S5 is a tool operation task where the robot shakes up and down the handle of a camera tripod in the collaboration mode to adjust the camera's orientation back and forth two times between poses (a) and (b) [see Fig. 6]. Unlike S3 and S4, which have stiff environments, S5 has an environment with zero stiffness, as the stiffness of the camera tripod can be omitted.

The proposed controller (8) with (3), (9), and (17) is applied to conduct experiments, in which we set the control parameters $K_c = \text{diag}(18, 18, 18, 18, 14.4, 14.4, 6)$, $\Lambda = \text{diag}(10, 10, 10, 10, 10, 5)$, $\Gamma = 0.5I$, and $k_a = 1$, $k_b = 3$, the weight parameters $Q = \text{diag}(0.35I, 5.5I)$, $\delta_1 = 1$, $\delta_2 = 5$, $\delta_3 = 2$, $\delta_4 = 3$, $\alpha_1 = 0.1$, $\alpha_2 = 1$, and $\beta = 1.5$, and the filter parameters $\lambda_1 = 12$ and $\lambda_2 = 25$. A composite learning constant impedance controller (CL-CIC) is obtained by simply setting $\omega_d = 1$ and $\omega_s = 0$ in (8) of the proposed CL-VIC for comparisons. Note that the CL-CIC has the same closed-loop dynamics as the robot-dominant mode in Section III-C after the convergence of the estimation error $\tilde{W}$.

Experimental results by the two controllers are demonstrated in Figs. 7 and 8, in which the parameter estimate norm $\|\tilde{W}\|$ converges to a certain constant under the IE condition before performing the interaction tasks, and it remains unchanged even during the execution of the interaction tasks [see Fig. 7(a)]. The usage of the weights $\omega_s = 0$ and $\omega_d = 1$ during the free motion (S1) [see Fig. 7(c), (d)] results in almost the same tracking errors e for the CL-VIC and CL-CIC [see Fig. 7(e)], which is consistent with the analysis of the robot-dominant mode in Section III-C. As the CL-VIC has low stiffness and damping (due to a large ω_s and a small ω_d) during contact [see Fig. 7(c), (d) and 8(c), (d)], it shows a larger e in the contact tasks (S3-S5) [see Figs. 7(f) and 8(b)], which is advantageous to keep the contact torque τ_e small in both the stiff and damping contact environments, avoiding damage to equipment [see Figs. 7(b) and 8(a)]. Besides, the robot transitions into the human-dominant mode when the

TABLE I
KEY EXPERIMENTAL DATA BY TWO CONTROLLERS IN TASKS S1-S5

Metric	Proposed CL-VIC	Baseline CL-CIC
t_c	33.0 s	33.0 s
$\ e\ _{\max}$ (S1)	0.02 rad	0.02 rad
$\ \tau\ _{\max}$ (S2)	12 Nm	-
$\ \tau\ _{\max}$ (S3)	7.1 Nm	10.5 Nm
$\ \tau\ _{\max}$ (S4)	8.0 Nm	13.8 Nm
$\ \tau\ _{\max}$ (S5)	7.4 Nm	10.15 Nm

human intervenes, as ω_s rapidly increases to 1 [see Fig. 7(c)]. Hence, the human can freely operate the robot with a safe τ_e [see Fig. 7(b), (c)]. Moreover, ω_d decreases as the robot velocity $\dot{q}$ increases [see Fig. 7(d)], which makes the robot respond faster to human instruction in the human-dominant mode.

Some key experimental data by the two methods in Tasks S1–S5 are explicitly provided in Table I, including the convergence time t_c of the parameter estimate $\tilde{W}$, the peak tracking error norm $\|e\|_{\max}$ in S1, and the peak contact torque norm $\|\tau\|_{\max}$ in S2–S5. These data quantitatively validate the following key merits of the proposed CL-VIC: 1) The IE condition can hold after a short learning process, which improves system robustness as exciting the robot in all tasks is not necessary; 2) the variable impedance strategy does not affect tracking accuracy in free motion; 3) the human can smoothly transition the robot into the human-dominant mode and safely interact with the robot; 4) it guarantees safety and robustness with different contact environments. Note that S2 is skipped for the CL-CIC, as it does not support the human-dominant mode. Besides, the tracking error e in S2 is not shown due to the complete compliance of the robot to the human in this mode.

To fairly evaluate the safety of the human-dominant mode by the proposed CL-VIC in Task S2, we provide test results of 10 subjects in Fig. 9, where the average of $\|\tau\|_{\max}$ is 11.01 N.m. In addition, $\|\tau\|_{\max}$ by the proposed CL-VIC in the human-dominant mode does not apparently increase with the peak joint velocity norm $\|\dot{q}\|_{\max}$, attributed to the decreasing damping with the increasing joint velocity $\dot{q}$.

V. CONCLUSION

This letter has proposed a novel CL-VIC strategy for robots to achieve variable target impedance under parametric uncertainty. The proposed control scheme ensures smooth transitions between the robot-dominant and human-dominant modes, and the composite learning mechanism ensures the exact estimation of the robot dynamics under the much weaker IE condition. Thus, the proposed method does not depend on dynamic model information and achieves safety and efficacy during interaction. Experimental results have validated that the 7-DoF collaborative robot with the proposed method transitions stably and smoothly between different modes and interacts safely with environments under different stiffness values. In the future, we would extend the proposed CL-VIC to FJR that explicitly consider underactuated phenomena, which requires some adaptations due to the underactuated feature of FJR.

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